

Generation of Visual Human Behaviour In Dyadic Scenarios

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Agenda

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- 2 Related Works
- 3 Limitations
- 4 Our Contributions
- 5 Approach

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- In Dyadic conversation, Conversational agents (e.g Robot) need to adapt their behaviour to their interlocutors (e.g human)
- There are many possible settings or task for example generating non-verbal behaviour for agents (e.g facial expressions only, or upper body postures) from multi-modal input
- Our setting is that we are trying to predict upper body gestures including facial expressions of two interactants from the speech of these interactants.

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- Earlier works where based on deterministic models, those models may generate non-vivid and repetitive gestures or even small range of gestures (eg.beat) depending on trained Dataset
- More Recent methods or modals are using probabilistic approaches to generate either full body gestures Habibie et al. [2] or only facial expressions Feng et al. [1], and Jonell et al. [4] from multi-modal input

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- Those previously mentioned works didn't model the semantics of the language from input speech
- Usually Facial expression and Gestures that co-occur with speech depend greatly on the semantic content
- There are also many factors that can affect the social interaction such as personal characteristics and interpersonal relationship of both interlocutors

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- Modeling the semantic content of the input speech for example using Automatic Speech Recognition deep learning model
- Representing those context factors in our data to train our model for example personality, Age, gender and the mood (Appendix)

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- We will use a probabilistic model based on the work proposed by Jonell et al. [4]¹ their model is adapted to MoGlowHenter, Alexanderson, and Beskow [3] which is basically a normalizing flows
- Normalizing flow is easier to use compared to other generative models such as GANs or AVE because we can evaluate the density which is important for our task

¹Patrik Jonell et al. “Let’s Face It: Probabilistic Multi-modal Interlocutor-aware Generation of Facial Gestures in Dyadic Settings”. In: *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents*. 2020, pp. 1–8.

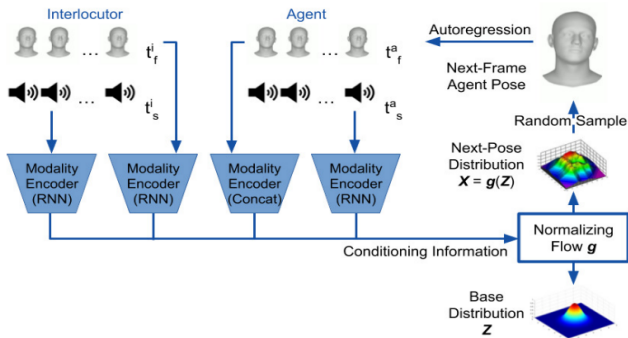
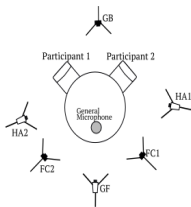


Figure 1: System Architecture



(a)



(b)

Figure 1. Recording environment. We used six tripod-mounted cameras, namely **GB**: General Rear camera, **GF**: General Frontal camera, **HA**: individual High Angle cameras and **FC**: individual Frontal Cameras, and two ego cameras **E** (one per participant, placed around their neck). a) Position of cameras, general microphone and participants. b) Example of the time-synchronized 8 views.



Figure 2. Examples of the 5 tasks included in the UDIVA dataset from 5 sessions. From left to right: *Talk*, *Lego*, *Animals*, *Ghost*, *Gaze*.

Figure 2

References I



Will Feng et al. “Learn2Smile: Learning non-verbal interaction through observation”. In: *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2017, pp. 4131–4138.



Ikhsanul Habibie et al. “Learning Speech-driven 3D Conversational Gestures from Video”. In: *arXiv preprint arXiv:2102.06837* (2021).



Gustav Eje Henter, Simon Alexanderson, and Jonas Beskow. “Moglow: Probabilistic and controllable motion synthesis using normalising flows”. In: *ACM Transactions on Graphics (TOG)* 39.6 (2020), pp. 1–14.

References II



Patrik Jonell et al. “Let’s Face It: Probabilistic Multi-modal Interlocutor-aware Generation of Facial Gestures in Dyadic Settings”. In: *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents*. 2020, pp. 1–8.



Cristina Palmero et al. “Context-Aware Personality Inference in Dyadic Scenarios: Introducing the UDIVA Dataset.”. In: *WACV (Workshops)*. 2021, pp. 1–12.

Table 2. Description of the different sources of context included as metadata in the proposed personality inference model.

Context type			Source	Value range normalization	Output size
Individual	Stable (across sessions)	Age	Self-reported	$[17, 75] \rightarrow [0, 1]$	1D
		Gender	Self-reported	$\{F, M\} \rightarrow \{0, 1\}$	1D
		Cultural background	Self-reported (country of origin)	Recategorization based on cultural differences [53]	6D (one-hot encoding)
	Transient (per session)	Session index	Session info.	$[1, 5] \rightarrow [0, 1]$	1D
		Pre-session mood	Self-reported [32] (8 categories*, Likert scale)	$[1, 5] \rightarrow [0, 1]$ (for each category)	8D
		Pre-session fatigue	Self-reported (Rating scale)	$[0^\dagger, 10] \rightarrow [0, 1]$	1D
Session	Order of the task within the session		Session info.	$[1, 4] \rightarrow [0, 1]$	1D
	Task difficulty [†]		External survey	$[0, 3] \rightarrow [0, 1]$	1D
Dyadic	Interlocutors* relationship		Self-reported	$\{N, Y\} \rightarrow \{0, 1\}$	1D

*Categories: *good*, *bad*, *happy*, *sad*, *friendly*, *unfriendly*, *tense*, and *relaxed*.

[†] Sessions with fatigue data missing were assigned a value of 0.

[‡] Tasks with no difficulty level associated were assigned a value of 0.